

## Disorders of Extracellular Volume

David H. Ellison

### EXTRACELLULAR FLUID COMPARTMENT

Water makes up 60% of a typical man's body weight and 50% of a woman's body weight. Total body water (TBW) is distributed in two compartments: the intracellular fluid (ICF) compartment (55%–65% of TBW) and extracellular fluid (ECF) compartment (35%–45% of TBW). The ECF is further subdivided into two spaces: the *interstitial* space, which accounts for about 75% of the ECF, and the *intravascular* space, which represents the remaining 25% (Fig. 8.1).

Water diffuses freely between the intracellular space and the extracellular spaces in response to gradients in effective osmolality, where effective osmolality (tonicity) is the product of the solute concentration and its reflection coefficient (similar to the inverse of permeability). Therefore, the amount of water in different compartments depends primarily on the quantity of effective osmoles in that compartment. The major cation in the ECF is sodium ion ( $\text{Na}^+$ ), and the major intracellular cation is potassium ion ( $\text{K}^+$ ); the number of cations always equals the number of anions in fluid. This uneven ion distribution is maintained by active transport through the  $\text{Na}^+/\text{K}^+$ -adenosine triphosphate (ATP)-dependent pumps on the cell membrane, which determines the relative volume of different compartments. Because sodium is the predominant extracellular cation, the ECF volume is determined primarily by the sodium content of the body. Total body sodium largely depends on salt intake and kidney excretion; the latter is tightly regulated.

Fluid movement between the intravascular and interstitial spaces of the ECF occurs across the capillary wall and is governed by Starling forces: the capillary hydrostatic pressure and colloid osmotic pressure. Unlike cell membranes, capillary membranes are highly permeable to small solutes, such as  $\text{Na}^+$ , rendering small molecules incapable of generating transcapillary water movement. Plasma proteins, especially albumin, play special roles in retaining fluid within capillaries because proteins are poorly permeable across capillary membranes. The outward transcapillary hydrostatic pressure gradient exceeds the corresponding inward oncotic pressure gradient, thereby favoring movement of plasma ultrafiltrate into the interstitial space.

Titze and colleagues<sup>1</sup> suggest that this model is oversimplified. They emphasize that not all fluid compartments are in osmotic equilibrium and not all extracellular sodium is osmotically active. Accordingly, sodium can be stored in compartments such as the skin (whether osmotically silent<sup>2</sup> or osmotically active<sup>3</sup> is not clear) and then released in a rhythmic manner that is independent of ECF volume.<sup>4</sup> The existence of such compartments has been recognized for many years; for

example, in cartilage, glycosaminoglycans attract sodium to generate a hypertonic environment into which water movement is resisted by rigid collagen,<sup>5</sup> and a “gel phase” containing glycosaminoglycans was incorporated into the well-known model generated by Guyton and colleagues.<sup>6</sup> Immune cells contribute to the skin sodium storage, which may help determine the ability of skin to resist microbial pathogens. Sodium homeostasis relies on the crucial role of the lymphatics in returning sodium to the circulation. Despite these important caveats to the concept of a uniform interstitial environment, treating edema removes fluid primarily from an “interstitial” (nonplasma) compartment, indicating that most extravascular fluid is in osmotic equilibrium with plasma.<sup>7</sup>

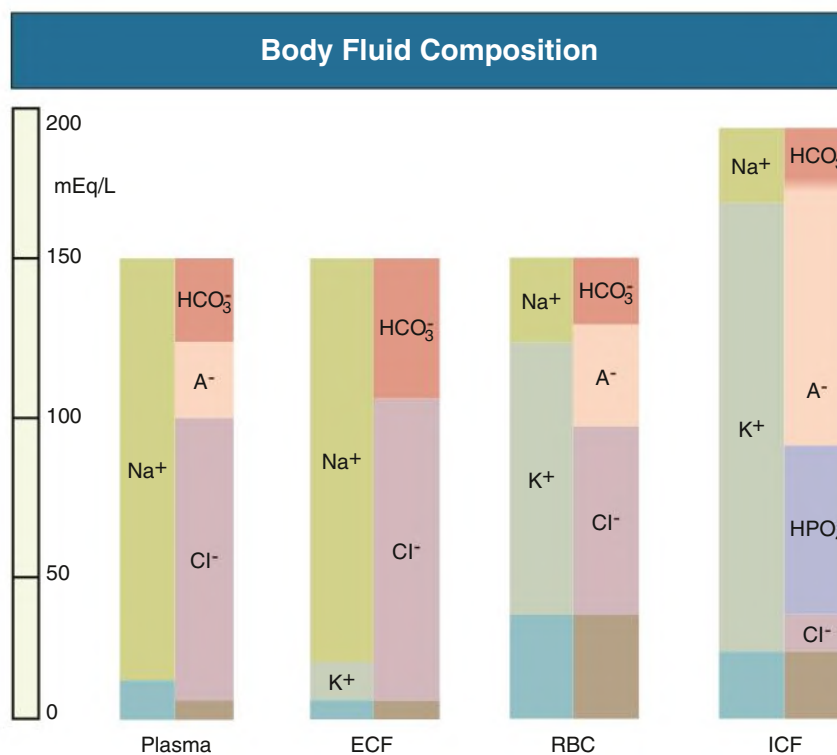
The ECF volume determines the adequacy of the circulation and, in turn, the adequacy of delivery of oxygen, nutrients, and other substances needed for organ functions; it is also necessary for removal of waste products. This is achieved despite day-to-day variation in the intake of sodium and water, with the ECF volume varying by only 1% to 2%.

### REGULATION OF EXTRACELLULAR FLUID HOMEOSTASIS

Circulatory stability depends on homeostatic mechanisms that include an *afferent* sensing limb, involving volume and stretch detectors distributed throughout the vascular bed, and an *efferent* effector limb (Table 8.1). Adjustments in the effector mechanisms occur in response to afferent stimuli by sensing-limb detectors to modify circulatory parameters. Disorders of either sensing mechanisms or effector mechanisms can lead to failure of adjustment of sodium handling by the kidney, with resultant hypertension or edema formation in the patient with positive sodium balance or hypotension and hypovolemia in the patient with negative sodium balance.

#### Afferent (Sensor) Limb

Afferent limb (sensing) sites include low-pressure cardiopulmonary receptors (atrial, ventricular, and pulmonary stretch receptors), high-pressure arterial baroreceptors (carotid, aortic arch, and kidney sensors), central nervous system (CNS) receptors, and hepatic receptors. The cardiac atria possess the distensibility and compliance needed to monitor changes in intrathoracic venous volume. Atrial distention and a sodium load cause release of atrial natriuretic peptide (ANP), a polypeptide normally stored in secretory granules within atrial myocytes. The closely related brain natriuretic peptide (BNP) is stored primarily



**Fig. 8.1** Gamblegram of Body Composition. A 70-kg man contains 42 L (60%) of water, whereas a 60-kg woman contains 36 L (50%) of water. Of water, two-thirds is intracellular (ICF) and one-third is extracellular (ECF). The osmolality in compartments is similar, even though solute concentrations differ, owing to valence of ions. *RBC*, Red blood cell; *A*, Other anions.

**TABLE 8.1 Homeostatic Mechanisms in Extracellular Fluid Volume**

| Afferent (Sensing)               | Efferent (Effector)                  |
|----------------------------------|--------------------------------------|
| Cardiopulmonary receptors        | Renin-angiotensin-aldosterone system |
| • Atrial                         | Prostaglandins                       |
| • Ventricular                    | Arginine vasopressin                 |
| • Pulmonary                      | Natriuretic peptides                 |
| High-pressure baroreceptors      | • Atrial                             |
| • Carotid                        | • Brain                              |
| • Aortic                         | • C-type                             |
| • Kidney                         | Other hormones                       |
| • Pressure sensors <sup>a</sup>  | • Nitric oxide                       |
| • Glomerular afferent            | • Endothelin                         |
| • Juxtaglomerular apparatus      | • Kallikrein-kinin system            |
| Central nervous system receptors |                                      |
| Hepatic receptors                |                                      |

<sup>a</sup>Pressure sensors are unspecified receptors contributing to pressure natriuresis.

in ventricular myocardium and is released when ventricular diastolic pressure rises. An increase in left atrial pressure also sends signals to the hypothalamus that can suppress the release of antidiuretic hormone (ADH), also called arginine vasopressin (AVP). These atrial-renal and atrial-hypothalamic reflexes enhance kidney sodium and water excretion on sensing of a distended left atrium.

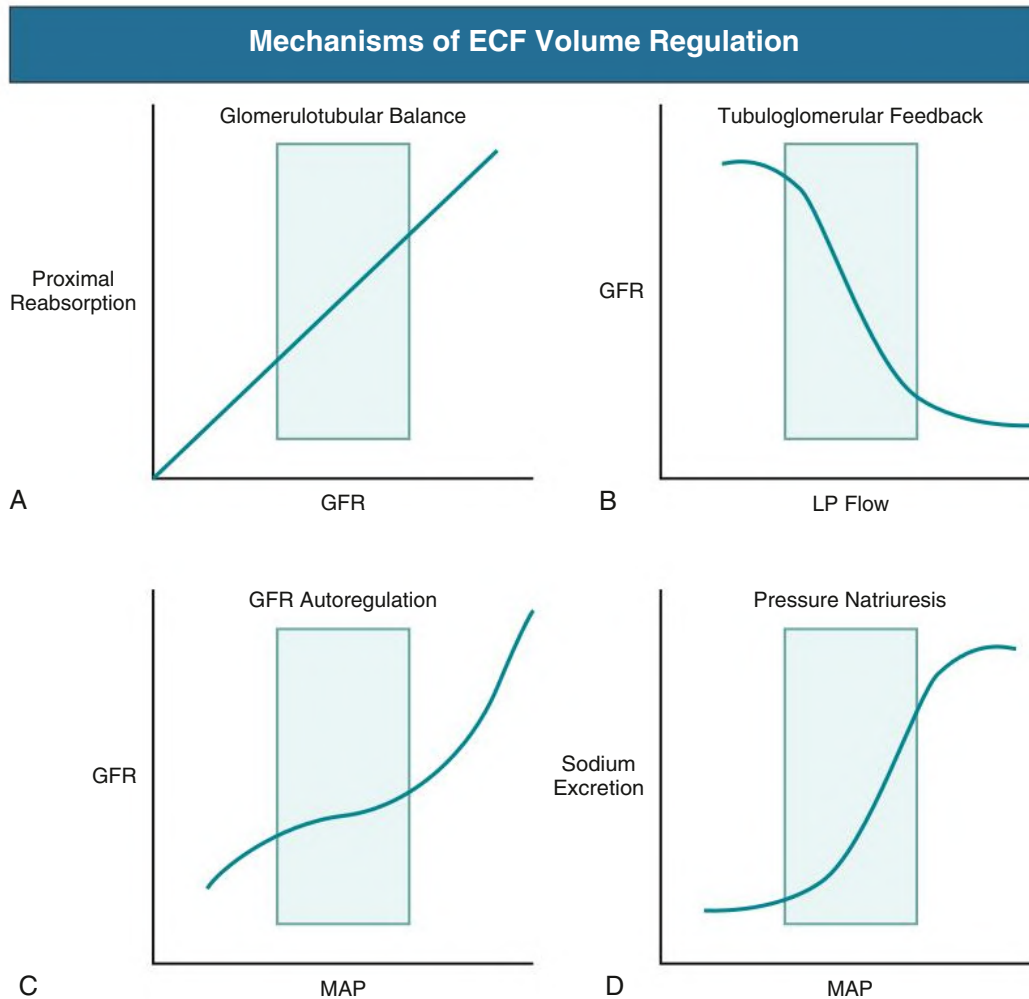
The sensitive arterial stretch receptors in the carotid artery and in the aortic arch respond to a decrease in arterial pressure. Information from these nerve endings is carried by the vagal and glossopharyngeal nerves to vasomotor centers in the medulla and brainstem. In a normal

situation, these receptors exert a tonic restraining effect on the heart and circulation by inhibiting the sympathetic outflow and augmenting parasympathetic activity. In addition, changes in transmural pressure across the arterial vessels and the atria also influence the secretion of AVP and renin and the release of ANP. Activation of the arterial receptors signals the kidney to retain sodium and water through increases in sympathetic activity and vasopressin release. Stimulation of the sympathetic nervous system (SNS) also enhances the renin-angiotensin-aldosterone system (RAAS). A rise in arterial pressure elicits the opposite response, resulting in decreased catecholamine release and natriuresis.

Kidney-sensing mechanisms include the juxtaglomerular apparatus (JGA), which is involved in the generation and release of renin from the kidney and in tubuloglomerular feedback (TGF) (see later discussion). Renin secretion is inversely related to perfusion pressure and directly related to intrarenal tissue pressure. Solute delivery to the macula densa is also an important determinant of renin release; an increase in sodium chloride entry into macula densa cells inhibits renin release, whereas a decrease in entry stimulates it. Kidney nerve stimulation through activation of  $\beta$ -adrenergic receptors of the JGA directly enhances renin release. Other receptors reside in the CNS and hepatic circulation but have been less well defined.

### Kidney Hemodynamics and Sodium Excretion

Increases or decreases in glomerular filtration rate (GFR) lead to parallel changes in NaCl reabsorption (Fig. 8.2), the phenomenon of glomerulotubular balance (GTB), so that GFR is not a major determinant of net solute excretion. A second process, TGF, senses NaCl at the macula densa to adjust GFR. High luminal NaCl concentration at the macula densa occurs during high loop segment flow rates and volume expansion. This leads to constriction of the nearby afferent arteriole.



**Fig. 8.2** Overall Mechanisms of Extracellular Fluid (ECF) Volume Regulation. (A) Proximal reabsorption rises with increased glomerular filtration rate (GFR) (glomerulotubular balance). (B) Relationship between late proximal (LP) flow and GFR (tubuloglomerular feedback). (C) GFR is autoregulated across mean arterial pressure (MAP). (D) Sodium excretion increases, as MAP increases, the pressure natriuresis. Boxes indicate typical operating ranges.

This process reduces GFR (see Fig. 8.2) and therefore proximal flow, tending to keep solute excretion rates constant. Although functionally independent of GTB, the sequential arrangement of GTB and TGF means they work in concert and are remarkably effective at maintaining NaCl excretion rates in the face of changing GFR.<sup>8</sup>

The importance of TGF for kidney health has been emphasized recently by the introduction of sodium glucose cotransporter 2 inhibitors. These drugs inhibit glucose and sodium reabsorption along the proximal tubule and were developed as treatments for diabetes mellitus. Nevertheless, they have shown a remarkable ability to slow the progression of chronic kidney disease (CKD), not only when caused by diabetes, but also from other causes.<sup>9</sup> Although the mechanisms are likely pleiotropic, they strongly activate TGF by delivering more sodium to the macula densa, thereby reducing intraglomerular pressure, an effect similar to that observed with angiotensin-converting enzyme (ACE) inhibitors.<sup>10</sup>

### Pressure Natriuresis

Both kidney blood flow and GFR are autoregulated (see Fig. 8.2), meaning they are relatively insensitive to variations in arterial pressure, within a range of typical pressure values. In contrast, urinary Na<sup>+</sup> excretion is strongly affected by even modest variations in pressure,

with a rise in pressure increasing kidney Na<sup>+</sup> excretion (see Fig. 8.2). This process appears to be intrinsic to the kidney, but the shape of the relationship is strikingly altered by the RAAS. This phenomenon, termed *pressure natriuresis*, likely results from the adjustment of Na<sup>+</sup> reabsorption by several segments of the nephron, via pathways that remain incompletely defined. The dominant importance of pressure natriuresis, at least under sodium retentive conditions, is clear from experimental models. Natriuretic sensitivity to natriuretic peptides in heart failure can be restored<sup>11</sup> by a rise in kidney perfusion pressure (RPP). Conversely, escape from the sodium retentive effects of aldosterone<sup>12</sup> or angiotensin II (Ang II)<sup>13</sup> also requires a rise in RPP.

### Efferent (Effector) Limb

The stimulation of the effector limb of the ECF volume homeostasis leads to activation of effector mechanisms (see Table 8.1). These effector mechanisms aim predominantly at modulation of kidney sodium and water excretion to preserve circulatory stability.

### Renin-Angiotensin-Aldosterone System

Renin secretion from the JGA increases in response to depletion of the ECF volume as a result of the processes described previously. Renin converts angiotensinogen to Ang I, which is then converted to Ang II

by the action of the ACE; Ang II can subsequently affect circulatory stability and ECF volume homeostasis. Ang II is a vasoconstrictor that stimulates sodium retention and aldosterone release, which all maintain arterial pressure when ECF volume is low. Ang II has complex effects on GFR and kidney plasma flow (RPF), but when the ECF volume is low, it preferentially increases kidney efferent arteriolar tone, thus tending to preserve GFR. Ang II also increases the filtration fraction by altering Starling forces across the glomerulus, which leads to enhanced proximal sodium and water retention.<sup>14</sup>

Ang II also augments sympathetic neurotransmission and enhances the TGF mechanism. In addition to these indirect mechanisms, Ang II directly enhances proximal tubular volume reabsorption by activating apical membrane sodium-hydrogen ( $\text{Na}^+\text{-H}^+$ ) exchangers. Ang II also enhances sodium absorption by stimulating aldosterone secretion, which, in turn, increases sodium reabsorption in the aldosterone-sensitive distal nephron.

### Sympathetic Nervous System

Sympathetic nerves that originate in the prevertebral celiac and paravertebral ganglia innervate cells of the afferent and efferent arterioles, JGA, and kidney tubule. Sympathetic nerves alter kidney sodium and water handling by direct and indirect mechanisms.<sup>15</sup> Increased nerve stimulation indirectly stimulates proximal tubular sodium reabsorption by altering preglomerular and postglomerular arteriolar tone, thereby influencing filtration fraction. Kidney nerves directly stimulate proximal tubular fluid reabsorption through receptors on the basolateral membrane of the proximal convoluted tubule cells. These effects on sodium handling are further amplified by the ability of the sympathetic nerves to stimulate renin release, which leads to the formation of Ang II and aldosterone.

### Natriuretic Peptides

ANP and BNP augment sodium and water excretion by increasing GFR, possibly by dilating the afferent arteriole and constricting the efferent arteriole. Furthermore, they inhibit sodium reabsorption in the cortical collecting tubule and inner medullary collecting duct, reduce renin and aldosterone secretion, and oppose the vasoconstrictive effects of Ang II.<sup>16</sup> Circulating levels of ANP and BNP are elevated in congestive heart failure (CHF) and in cirrhosis with ascites but do not overcome the sodium-retaining effects of low RPP.

### Prostaglandins

Prostaglandins are derived from arachidonic acid and modulate kidney blood flow and sodium handling. Important kidney prostaglandins include prostaglandin  $\text{I}_2$ , which mediates baroreceptor (but not  $\beta$ -adrenergic) stimulation of renin release. Prostaglandin  $\text{E}_2$  is stimulated by Ang II and has vasodilatory properties. Increased levels of Ang II, AVP, and catecholamines stimulate synthesis of prostaglandins, which, in turn, act to dilate the kidney vasculature, inhibit sodium and water reabsorption, and stimulate renin release. In situations of ECF or effective arterial blood volume (EABV) depletion, the combination of prostaglandin-mediated afferent arteriolar dilation and Ang II-mediated efferent arteriolar constriction plays a central role in autoregulatory maintenance of GFR. Often, in these situations, interference with efferent vasoconstriction (e.g., ACE inhibition), and with afferent vasodilation (e.g., nonsteroidal antiinflammatory drugs [NSAIDs]) leads to a precipitous decline in GFR, manifested as acute kidney injury (AKI).

### Arginine Vasopressin

The polypeptide AVP is synthesized in supraoptic and paraventricular nuclei of the hypothalamus and is secreted by the posterior pituitary gland. Hypertonicity is the predominant stimulus for AVP release

under typical conditions. Substantial reductions in EABV, however, act as a second, nonosmotic regulatory pathway.<sup>17</sup> AVP release is suppressed in response to ECF volume overload sensed by increased afferent impulses from arterial baroreceptors and atrial receptors, whereas decreased ECF volume has the opposite effect. AVP release leads to antidiuresis via  $\text{V}_2$  receptors and, at higher concentrations, to systemic vasoconstriction through the  $\text{V}_1$  receptors.<sup>18</sup> The antidiuretic action of AVP results from the effect on the principal cell of the collecting duct through activation of the  $\text{V}_2$  receptor. AVP increases the synthesis and provokes the insertion of aquaporin 2 (AQP2) water channels into the luminal membrane, thereby allowing water to be reabsorbed down the favorable osmotic gradient generated by the countercurrent multiplier system. AVP also enhances  $\text{Na}^+$  reabsorption via the epithelial sodium channel, ENaC, and thereby  $\text{K}^+$  secretion. AVP appears to have synergistic effects with aldosterone on sodium transport in the cortical collecting duct.<sup>19</sup> AVP stimulates potassium secretion by the distal nephron, and this preserves potassium balance during ECF depletion, when circulating levels of vasopressin are high and tubular delivery of sodium and fluid is reduced. It should be noted, however, that states of excess AVP secretion are manifested primarily by excess water, and not salt, accumulation, because of offset physiologic control mechanisms.

### Other Hormones

Other hormones that contribute to kidney sodium handling and ECF volume homeostasis include nitric oxide (NO), endothelin, and the kallikrein-kinin system. NO is an endothelium-derived mediator that participates in the natriuretic responses to increases in blood pressure or ECF volume expansion. Endothelins, in addition to their potent actions to constrict vascular smooth muscle, are natriuretic factors. Endothelin 1, via the endothelin B ( $\text{ET}_\text{B}$ ) receptor, increases NO production, which tends to increase urinary salt excretion. Kinins are potent vasodilator peptides, but their physiologic roles are not yet fully defined.

### Terms Useful for Disorders of Extracellular Fluid Volume

In clinical practice, it is helpful to view disorders of ECF volume as distinct from disorders of water balance. The latter reflect gain or loss of electrolyte-free water, typically resulting in changes in plasma osmolality (hyponatremia or hypernatremia); when losses or gains are substantial, detectable changes in ECF volume can occur. Changes in ECF volume, however, often termed *volume contraction* or *volume expansion*, reflect gain or loss of NaCl. Water is often retained or lost secondarily.

Loss of salt and water produces true depletion of ECF volume, but there are many pathologic conditions (e.g., heart failure [HF]; cirrhosis) in which ECF volume is expanded but the kidneys behave as if it is contracted. To account for this discrepancy, the term EABV is used to describe the blood volume detected by the sensitive arterial baroreceptors. The EABV can change independently of the total ECF volume. The state of the EABV is often inferred from the behavior of the kidneys.

### EXTRACELLULAR FLUID VOLUME CONTRACTION

Contraction of ECF volume typically results from sodium losses that exceed intake. Losses may be in the kidney or extrarenal through the gastrointestinal (GI) tract, skin, and lungs or by sequestration in potential spaces in the body (e.g., abdomen, muscle) that are not in hemodynamic equilibrium with the ECF (Table 8.2). The reduction in ECF volume occurs from both the interstitial and intravascular compartments. The loss of solute-free water (e.g., diabetes insipidus [DI]) has a lesser effect on intravascular volume because water is lost from

**TABLE 8.2 Major Causes of Extracellular Fluid Volume Depletion**

| Kidney                                                                                                                                                                                                                                                                                                                                                                                                       | Extrarenal                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Diuretic use                                                                                                                                                                                                                                                                                                                                                                                                 | Gastrointestinal losses                                                                                                                |
| Tubular disorders                                                                                                                                                                                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>• Vomiting</li> <li>• Gastrointestinal suctioning</li> </ul>                                    |
| <ul style="list-style-type: none"> <li>• Genetic <ul style="list-style-type: none"> <li>• Bartter and Gitelman syndromes</li> <li>• Pseudohypoaldosteronism type 1</li> </ul> </li> <li>• Acquired tubular disorders <ul style="list-style-type: none"> <li>• Acute kidney injury</li> <li>• Recovery phase of oliguric kidney injury</li> <li>• Release of urinary tract obstruction</li> </ul> </li> </ul> | Diarrhea <ul style="list-style-type: none"> <li>• Ileostomy and colostomy secretions</li> </ul>                                        |
| Hormonal and metabolic disturbances                                                                                                                                                                                                                                                                                                                                                                          | Dermal losses                                                                                                                          |
| <ul style="list-style-type: none"> <li>• Mineralocorticoid deficiency or resistance <ul style="list-style-type: none"> <li>• Primary adrenal insufficiency</li> <li>• Hyporeninemic hypoaldosteronism</li> <li>• Diabetes mellitus</li> <li>• Chronic interstitial kidney diseases</li> </ul> </li> <li>• Solute diuresis</li> </ul>                                                                         | <ul style="list-style-type: none"> <li>• Sweat</li> <li>• Exudative skin disease</li> </ul>                                            |
| Kidney water loss                                                                                                                                                                                                                                                                                                                                                                                            | Third-space sequestration                                                                                                              |
| <ul style="list-style-type: none"> <li>• Diabetes insipidus</li> </ul>                                                                                                                                                                                                                                                                                                                                       | <ul style="list-style-type: none"> <li>• Ascites</li> <li>• Pleural effusion, hydrothorax</li> <li>• Intestinal obstruction</li> </ul> |
|                                                                                                                                                                                                                                                                                                                                                                                                              | Retroperitoneal collection                                                                                                             |
|                                                                                                                                                                                                                                                                                                                                                                                                              | Hemorrhage                                                                                                                             |
|                                                                                                                                                                                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>• Internal</li> <li>• External</li> </ul>                                                       |

all aqueous spaces in the body, and the body's solute content remains unchanged. Thus, in this case of disordered water balance, hypertonicity—rather than ECF volume contraction—predominates.

## Extrarenal Causes

### Gastrointestinal Losses

Approximately 3 to 6 L of fluids and digestive juices are secreted daily throughout the GI tract, and most of this fluid is reabsorbed along more distal portions. Vomiting or nasogastric suction may cause volume loss that is usually accompanied by metabolic alkalosis, whereas diarrhea may result in volume depletion that is accompanied by metabolic acidosis. This results from abnormal loss of bicarbonate or salts of organic anions in the stool. An exception to this general principle is the rare syndrome known as congenital chloride-wasting diarrhea, in which, because of mutations in a protein that exchanges chloride for bicarbonate in the colon,<sup>20</sup> infants typically present with diarrhea and metabolic alkalosis.

### Dermal Losses

Sweat is typically hypotonic, leading to more water loss than salt loss. Sweat production can be excessive in high ambient temperature or with prolonged exercise in hot, humid climates and may lead to volume depletion. Loss of the skin barrier with superficial burns and exudative skin lesions may lead to significant ECF volume depletion.

### Third-Space Sequestration

Body fluid accumulation in potential spaces that are not in hemodynamic equilibrium with the ECF compartment can cause volume depletion. This pathologic accumulation, often called third-space

sequestration, includes ascites, hydrothorax, and intestinal obstruction, with fluid collecting in the peritoneal cavity, pleural space, and intestines, respectively, and leading to significant ECF volume loss. Severe pancreatitis may result in retroperitoneal fluid collections.

### Hemorrhage

Hemorrhage occurring internally (e.g., from bleeding esophageal varices) or externally (e.g., trauma) may lead to significant volume loss.

### Kidney Losses

In health, approximately 25,000 mmol of sodium is filtered every day. The small quantities of sodium excreted in urine relative to the filtered load depend on intact tubular reabsorptive mechanisms to adjust urinary sodium excretion to maintain ECF homeostasis. Impairment in the integrity of these mechanisms can result in significant volume depletion.

### Diuretic Use

Most of the widely used diuretic medications inhibit sodium transport pathways along the nephron (see later discussion). Diuretics may cause sodium wasting by the kidney with volume contraction and metabolic acid-base disturbances.

### Genetic and Acquired Tubular Disorders

Tubular sodium reabsorption may be disrupted in several genetic disorders that include Bartter syndrome and Gitelman syndrome; these are often called “tubulopathies.” They are caused by mutations of sodium, potassium, or chloride transporters or in regulatory factors. Although they result in sodium wasting and volume contraction, they typically present with hypokalemic metabolic alkalosis. They are discussed in more detail in [Chapter 49](#). Pseudohypoaldosteronism type 1 (PHA1) is another rare inherited disorder characterized by sodium wasting and hyperkalemic metabolic acidosis and is caused by mutations in either the epithelial sodium channel (ENaC) or the mineralocorticoid receptor.

Acquired tubular disorders that may be accompanied by salt wasting include AKI, during the recovery phase of oliguric AKI, or urinary obstruction (see [Chapters 61 and 74](#)). Sjögren syndrome may cause a phenotype resembling Gitelman syndrome, as may treatment with cisplatin.

### Hormonal and Metabolic Disturbances

Mineralocorticoid deficiency and resistance states often lead to sodium wasting. This may occur in the setting of primary adrenal insufficiency (Addison disease) and PHA1. Salt wasting also can be seen in chronic tubular and interstitial kidney diseases. Severe hyperglycemia or high levels of blood urea during release of urinary tract obstruction can lead to obligatory kidney sodium and water loss secondary to glycosuria or urea diuresis, respectively.

### Water Loss via the Kidney

DI represents a spectrum of diseases resulting from AVP deficiency (central DI) or tubular resistance to AVP (nephrogenic DI). The most common causes of polyuria from nephrogenic DI in adults are chronic lithium ingestion, hypercalcemia, and, less frequently, hypokalemia (see [Chapter 9](#)). In these disorders the tubular reabsorption of solute-free water is impaired. This generally results in a lesser effect on ECF volume because, in contrast to sodium, there is a relatively smaller amount of the TBW in the ECF compartment compared with the ICF compartment.



### BOX 8.1 Clinical Evaluation of Extracellular Fluid Volume Depletion

#### Mild to Moderate Volume Loss

- Thirst
- Delay in capillary refill
- Postural dizziness, weakness
- Dry mucous membranes and axillae
- Cool, clammy extremities and collapsed peripheral veins
- Tachypnea
- Tachycardia with pulse rate >100 beats/min or postural pulse increment of 30 beats/min or more
- Postural hypotension (systolic blood pressure decrease >20 mm Hg on standing)
- Low jugular venous pulse
- Oliguria

#### Severe Volume Loss and Hypovolemic Shock

- Depressed mental status (or loss of consciousness)
- Peripheral cyanosis
- Reduced skin turgor (in young patients)
- Marked tachycardia, low pulse volume
- Supine hypotension (systolic blood pressure <100 mm Hg)

### Clinical Manifestations of Extracellular Fluid Volume Contraction

The spectrum of the clinical manifestations of volume contraction (Box 8.1) depends on the amount and rate of ECF volume loss and on the vascular and kidney responses to that loss. The history and physical examination are usually the key factors in evaluating the presence and causes of hypovolemia because laboratory testing may appear normal. Symptoms are usually nonspecific and can range from mild postural symptoms, thirst, muscle cramps, and weakness to drowsiness and disturbed mentation with profound volume loss. Physical examination may reveal tachycardia, cold clammy skin, postural or recumbent hypotension, and reduced urine output, depending on the degree of volume loss. Low jugular venous pressure (JVP;  $\leq 5$ ) is consistent with volume depletion. However, the JVP may be elevated in patients with pulmonary hypertension or when the EABV is low. The lack of symptoms or discernible physical findings does not preclude volume depletion, and hemodynamic monitoring and administration of a fluid challenge may be necessary.

### Laboratory Tests

Hemoconcentration and increased serum albumin concentration may be seen early with hypovolemia, but anemia or hypoalbuminemia caused by a concomitant disease may confound interpretation of these laboratory values. In healthy individuals, the ratio of blood urea nitrogen (BUN) to serum creatinine is approximately 10:1 to 20:1 (measured in milligrams per deciliter). In volume-contracted states, this ratio may increase because of an associated differential increase in urea reabsorption in the collecting duct. Several clinical conditions affect this ratio. Upper GI tract hemorrhage and administration of corticosteroids increase urea production, and hence the ratio of BUN to creatinine increases. Malnutrition and liver disease diminish urea production, making the ratio less helpful.

Urine osmolality and specific gravity may be elevated in hypovolemic states but may be altered by an underlying kidney disease that leads to kidney sodium wasting, concomitant intake of diuretics, or a

solute diuresis. Hypovolemia normally promotes avid kidney sodium reabsorption, resulting in low urine sodium concentration and low fractional excretion of sodium. Urine chloride follows a similar pattern because sodium and chloride are generally reabsorbed together. Volume depletion with metabolic alkalosis (e.g., with vomiting) is an exception because of the need to excrete the excess bicarbonate in conjunction with sodium to maintain electroneutrality; in this case, urine chloride concentration is a better index of ECF volume contraction. The fractional excretion of sodium ( $FE_{Na}$ ) is calculated by the following formula:

$$FE_{Na} = [U_{Na} \times P_{creat} / U_{creat} \times P_{Na}] \times 100$$

where  $U_{Na}$  and  $U_{creat}$  are urinary sodium and creatinine concentrations, respectively, and  $P_{Na}$  and  $P_{creat}$  are plasma sodium and creatinine concentrations, respectively. In an oliguric patient with AKI,  $FE_{Na}$  less than 1% is consistent with volume depletion;  $FE_{Na}$  greater than 1% is more consistent with acute tubular necrosis, although the sensitivity and specificity of this test are limited.<sup>21</sup>

### Therapy of Extracellular Fluid Volume Contraction

The goal of treatment is to replace the fluid deficit and ongoing losses with a fluid that resembles the lost fluid. The first step is to determine the urgency of the hypovolemic condition. In severe hypovolemia or hypovolemic shock, immediate treatment should be initiated, typically with 1 to 2 L of isotonic crystalloid. The adequacy of repletion can then be monitored clinically with central venous pressure monitoring or, in the intensive care unit (ICU), with respiratory variation in the arterial pressure tracing. Estimating the magnitude of volume deficit is imprecise; thus a key component of successful treatment is frequent monitoring and adjustment of therapy.

Mild volume contraction usually can be corrected orally. It is important to note, however, that unlike oral rehydration solutions used for childhood diarrhea (50–90 mmol/L sodium), sports drinks typically contain very little NaCl (7–20 mmol/L) and do not replace substantial sodium losses, although normal kidneys can typically adjust for the discrepancy when losses are mild.

Crystalloid solutions (isotonic or slightly hypotonic) with sodium as the principal cation are effective because they distribute primarily in the ECF. One-third of an infusate of isotonic saline (0.9% NaCl) remains in the intravascular compartment, whereas two-thirds distributes into the interstitial compartment. Colloid-containing solutions include human albumin (5% and 25%) and hetastarch (6% hydroxyethyl starch [HES]), which remain within the vascular compartment (provided the transcapillary barrier is intact and not disrupted by capillary leak states such as often occurs with multiorgan failure). The solutions augment the plasma oncotic pressure and thus expand the plasma volume by counteracting the capillary hydraulic pressure.

Colloid-containing solutions have not shown an advantage in the treatment of hypovolemic states. A large multicenter trial that randomized medical and surgical critical patients to receive fluid resuscitation with 4% albumin or normal saline showed similar mortality, morbidity, and hospitalization rates in the two groups.<sup>22</sup> Another study randomly assigned ICU patients with severe sepsis to fluid resuscitation with either 6% HES or Ringer acetate solution.<sup>23</sup> Patients who received HES had increased mortality and were more likely to receive kidney replacement therapy. Consequently, artificial colloids should be avoided in patients with severe sepsis or at risk for developing AKI.<sup>24</sup>

Isotonic saline or a balanced salt solution (e.g., lactated Ringer's solution) is usually the preferred initial choice in volume-depleted

patients with normal serum  $[Na^+]$  and most of those with low serum  $[Na^+]$ . Furthermore, isotonic saline is the preferred fluid to restore ECF volume in hypovolemic patients with hypernatremia. Once euvolemia is established, hypotonic (0.45% NaCl) saline should be delivered to gradually correct tonicity. Administration of large volumes of isotonic saline may result in the development of hyperchloremic metabolic acidosis or AKI. Although a smaller randomized trial of patients admitted to the ICU did not detect a difference in AKI,<sup>25</sup> a recent large pragmatic trial of critically ill patients found that the use of balanced salt solutions was associated with an improved composite outcome of death, new kidney replacement therapy, or persistent kidney dysfunction compared with subjects receiving isotonic saline.<sup>26</sup>

Hypovolemic shock may be accompanied by lactic acidosis resulting from tissue hypoperfusion. Fluid resuscitation restores tissue oxygenation and will decrease the production of lactate, but lactated Ringer solution is inappropriate because the infused lactate will not be converted to bicarbonate. Correction of acidosis with sodium bicarbonate ( $NaHCO_3$ ) has the potential for increasing tonicity, expanding volume, worsening intracellular acidosis from increased carbon dioxide production, and not improving hemodynamics compared with isotonic saline. Whether  $NaHCO_3$  effectively corrects the impaired cardiac contractility associated with lactic acidosis has not been well documented by clinical studies. Therefore,  $NaHCO_3$  to manage lactic acidosis in the setting of volume depletion is often avoided unless the arterial pH is less than 7.1.

## EXTRACELLULAR FLUID VOLUME EXPANSION

Expansion of ECF volume usually results from kidney sodium and water retention. Generalized edema results from an apparent increase in the interstitial fluid volume, most often in response to HF, cirrhosis with ascites, and nephrotic syndrome. Weight gain of several kilograms usually precedes clinically apparent edema. Localized excess fluid may accumulate in the peritoneal and pleural cavities, leading to ascites and pleural effusion, respectively.

### Pathogenesis

Kidney sodium and water retention secondary to arterial underfilling leads to an alteration in capillary hemodynamics that favors fluid movement from the intravascular compartment into the interstitium. In general, these two processes account for edema formation.

### Capillary Hemodynamic Disturbances

According to the Starling equation, the exchange of fluid between the plasma and the interstitium is determined by the hydrostatic and oncotic pressures in each compartment. Interstitial fluid excess results from a decrease in plasma oncotic pressure or an increase in capillary hydrostatic pressure. In other words, edema is a result of an increase in fluid movement from the intravascular compartment to the interstitial space or a decrease in fluid movement from the interstitial space to the intravascular compartment, or both. Thus, the degree of interstitial fluid accumulation as determined by rate of fluid removal by the lymphatic vessels is a determinant of edema.

The capillary hydrostatic pressure is relatively insensitive to alterations in arterial pressure. The stability of the capillary pressure is a result of variations in the precapillary sphincter, which governs how much arterial pressure is transmitted to the capillary, a locally controlled response called *autoregulation*. In contrast, the venous end is not similarly well regulated. Therefore, when the blood volume expands, as in HF and kidney disease, capillary hydrostatic pressure increases and edema ensues. Venous obstruction works by the same

**TABLE 8.3 Major Causes of Extracellular Fluid Volume Expansion**

| Primary Kidney Sodium Retention | Secondary Kidney Sodium Retention <sup>a</sup> |
|---------------------------------|------------------------------------------------|
| Acute kidney injury             | Cardiac failure                                |
| Advanced chronic kidney disease | Cirrhosis                                      |
| Glomerular diseases             | Nephrotic syndrome                             |
|                                 | Idiopathic edema                               |
|                                 | Drug-induced edema                             |
|                                 | Pregnancy                                      |

<sup>a</sup>Secondary to reduced effective arterial blood volume depletion (arterial underfilling).

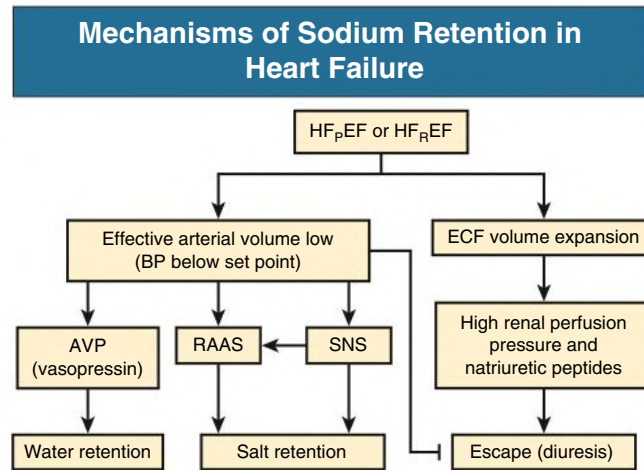
mechanism to cause edema, as exemplified by ascites formation in liver cirrhosis and by acute pulmonary edema after sudden impairment in cardiac function (e.g., myocardial infarction). In hepatic cirrhosis and nephrotic syndrome, another factor in edema formation is reduction in plasma oncotic pressure, with fluid transudation into the interstitial space. Even normal conditions favor net filtration into the interstitium because capillary hydrostatic pressure exceeds the plasma colloid pressure in several tissues throughout the capillary. In these tissues, a substantial amount of filtered fluid is returned to the circulation through lymphatic channels, which minimizes edema formation.

### Kidney Sodium Retention

The mechanism for maintenance of ECF volume expansion and edema formation is kidney sodium retention, which can be primary or secondary in response to reduction in EABV (Table 8.3).

**Primary sodium retention by the kidney.** A primary defect in kidney sodium excretion can occur with AKI, CKD, or glomerular disease. Patients with AKI have limited ability to excrete sodium and water. Advanced CKD may lead to sodium and water retention by GFR reduction, although in the absence of proteinuria, frank edema is uncommon. Primary sodium retention by the kidney characterizes some forms of glomerulonephritis and occurs through incompletely understood mechanisms in the presence of a relatively suppressed RAAS but frequently with decreased GFR. Filtered proteases may cleave and activate the sodium channel, ENaC, in these situations.<sup>27</sup>

Mineralocorticoid excess or enhanced mineralocorticoid activity are associated with sodium retention. However, because of mineralocorticoid escape (discussed previously), the clinical manifestation is generally hypertension rather than hypervolemia. In normal individuals, administration of a high-dose mineralocorticoid initially increases kidney sodium retention so that ECF volume is increased. However, kidney sodium retention then ceases, spontaneous natriuresis ensues, sodium balance is reestablished, and there is no detectable edema. This escape from mineralocorticoid-mediated sodium retention explains why edema is not a characteristic feature of primary hyperaldosteronism. The pathophysiologic mechanism of mineralocorticoid escape involves an increase in GFR and reduction of proximal tubular sodium and water reabsorption. This leads to an increase in sodium and water delivery to the distal nephron site of aldosterone action, which overrides the sodium reabsorption of aldosterone. Other contributing mechanisms include decreased expression of distal tubular thiazide-sensitive NaCl cotransporters,<sup>28</sup> increased secretion of ANP,<sup>29</sup> and pressure natriuresis. Regardless of the mechanisms involved, RPP must rise to enable the escape process<sup>12</sup>; because this can occur during primary kidney salt retention, these states are typically characterized by hypertension and not edema.



**Fig. 8.3 Mechanisms of Sodium Retention in Heart Failure.** Heart failure, from reduced ejection fraction (HF<sub>R</sub>EF) or preserved ejection fraction (HF<sub>p</sub>EF), activates neurohormonal systems, which lead to sodium and water retention, including the renin-angiotensin-aldosterone system (RAAS), the sympathetic nervous system (SNS), and arginine vasopressin (AVP). Ensuing ECF volume expansion increases natriuretic peptide secretion. Low kidney perfusion pressure (blood pressure below set point) causes resistance to natriuretic peptides and prevents escape from sodium and water retention from aldosterone, angiotensin II, and AVP.

**Secondary sodium retention by the kidneys.** An estimated 85% of blood circulates on the low-pressure venous side of the circulation, and 15% in the high-pressure arterial circulation. Thus, an increase in total blood volume could occur, even when there is underfilling of the arterial circulation, if the increase in total blood volume is primarily caused by expansion of the venous compartment. Underfilling of the arterial circulation could result from a decrease in cardiac output, as occurs in low-output cardiac failure, or from systemic arterial vasodilation, which occurs early in cirrhosis as a result of decreased systemic vascular resistance (SVR) in the splanchnic circulation.<sup>30</sup> Because arterial pressure is determined by the product of cardiac output and SVR, both states would be characterized by a decline in arterial pressure below the kidney's set point. This model proposes that kidney sodium retention triggered by arterial underfilling (arterial pressure less than pressure set point) is a compensatory response necessary to restore arterial circulatory integrity.

If there is arterial underfilling from decreased cardiac output or systemic arterial vasodilation, the hypotension is sensed by the arterial stretch receptors. This leads to activation of the efferent limb of body fluid volume homeostasis. Specifically, a decrease in glossopharyngeal and vagal tone from the carotid and aortic receptors to the CNS leads to a rapid increase in sympathetic activity with associated activation of the RAAS axis. If it is severe enough, the underfilling also leads to nonosmotic release of vasopressin. Additionally, a decrease in pressure at the kidney baroreceptors and decreased NaCl delivery to the macula densa increase renin secretion and thereby Ang II and aldosterone. The resultant increase in SVR and kidney sodium and water retention attenuates the arterial underfilling, through the Frank-Starling mechanism, and tends to restore arterial perfusion. Together, these actions maintain the arterial circulatory integrity and restore perfusion to vital organs at the expense of expanded ECF volume and edema.

### Sodium and Water Retention in Heart Failure

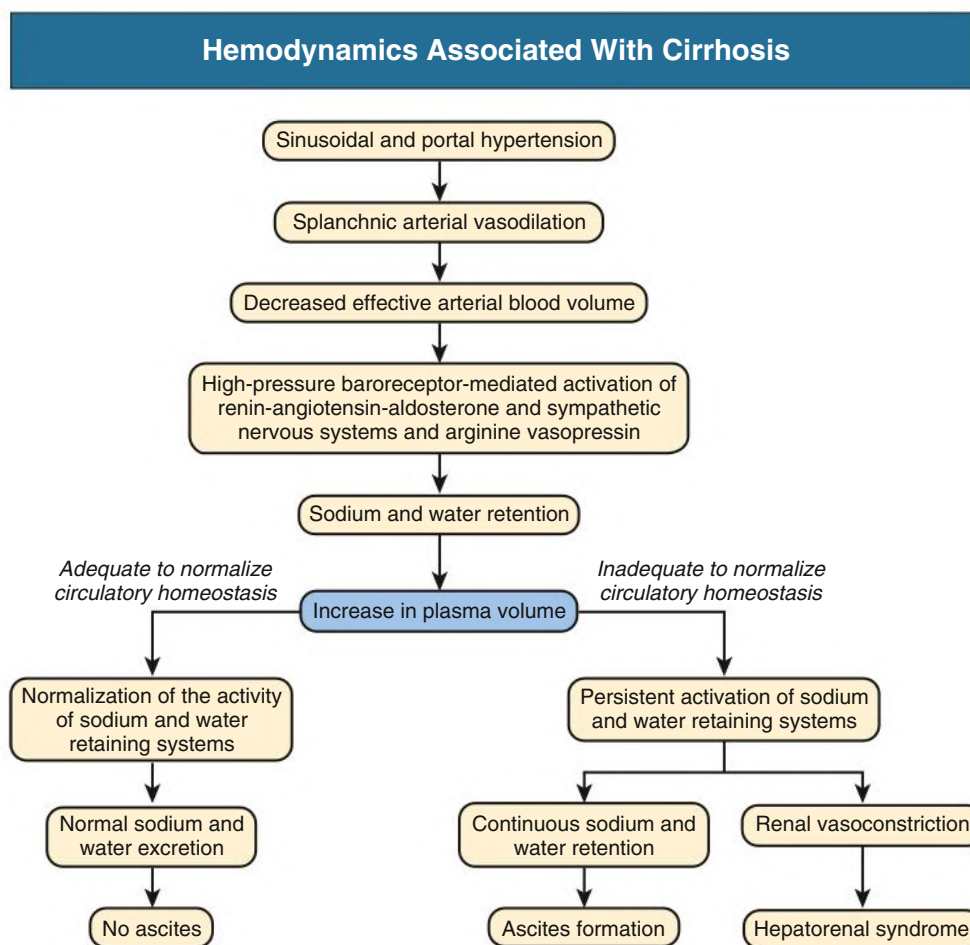
HF can occur with reduced or preserved ejection fraction (HF<sub>R</sub>EF and HF<sub>p</sub>EF, respectively), but the mechanisms involved in kidney sodium and water retention appear to involve similar mediators.<sup>31</sup> Decreased cardiac output with arterial underfilling leads to reduced stretch of arterial baroreceptors and reduced perfusion pressure in the kidney. In the case of HF<sub>p</sub>EF, this often involves a decrease below a hypertensive

baseline or below a baseline raised by CKD. This results in increased sympathetic discharge from the CNS and activation of the RAAS. Adrenergic stimulation and increased Ang II activate receptors on the proximal tubular epithelium that enhance sodium reabsorption. The kidney vasoconstriction of the glomerular efferent arteriole by Ang II in HF also alters net Starling forces in the peritubular capillary in a direction to enhance sodium reabsorption.<sup>32</sup> Thus, Ang II and  $\alpha$ -adrenergic stimulation increase sodium reabsorption in the proximal tubule by a direct effect on the proximal tubule epithelium and secondarily by kidney vasoconstriction. This subsequently leads to decreased sodium delivery to the collecting duct, which impairs escape from both aldosterone and natriuretic peptides.<sup>11,33</sup> This failure of escape explains why sodium retention and ECF volume expansion occur in HF (Fig. 8.3). Accordingly, patients with HF may have substantial natriuresis when spironolactone, a competitive mineralocorticoid receptor antagonist (MRA), is given in adequate doses to compete with increased endogenous aldosterone levels.<sup>34</sup>

The atrial-renal reflexes, which normally enhance kidney sodium excretion, are also impaired because plasma levels of ANP do not increase further when patients with dilated cardiomyopathy and mild HF receive a saline load, and the natriuretic response is also blunted. Autonomic dysfunction and blunted arterial baroreceptor sensitivity in HF are associated with increased circulating catecholamines and increased kidney sympathetic activity. There is also evidence for parasympathetic withdrawal in HF, in addition to the increase in sympathetic drive.

Another outcome of the neurohumoral activation that occurs in HF is the baroreceptor-mediated nonosmotic release of AVP.<sup>35</sup> This nonosmotic AVP stimulation overrides the osmotic regulation of AVP and is the major factor leading to the hyponatremia associated with end-stage HF.<sup>36</sup> AVP causes antidiuresis by activating V<sub>2</sub> receptors on the basolateral surface of the principal cells in the collecting duct.<sup>37</sup> Activation of these receptors initiates a cascade of intracellular signaling events by means of the adenylyl cyclase–cyclic adenosine monophosphate (cAMP) pathway, leading to an increase in AQP2 water channel protein expression and its trafficking to the apical membrane of the collecting duct. This sequence of events leads to increased water reabsorption and can cause hyponatremia, which is an ominous prognostic indicator in patients with HF.<sup>38</sup> Concurrently, increased nonosmotic AVP release stimulates V<sub>1</sub> receptors on vascular smooth muscle





**Fig. 8.4** Pathogenesis of Functional Kidney Abnormalities and Ascites Formation in Cirrhosis. (Modified from Ginés P, Cardenas A, Sola E, Schrier RW. Liver disease and the kidney. In: Coffman TM, Falk RJ, Molitoris BA, et al, eds. *Schrier's Diseases of the Kidney*. Wolters Kluwer; 2012:1965–1996.)

cells and thereby may increase SVR. This adaptive vasoconstrictive response may become maladaptive and contribute to cardiac dysfunction in patients with severe HF.

### Sodium and Water Retention in Cirrhosis

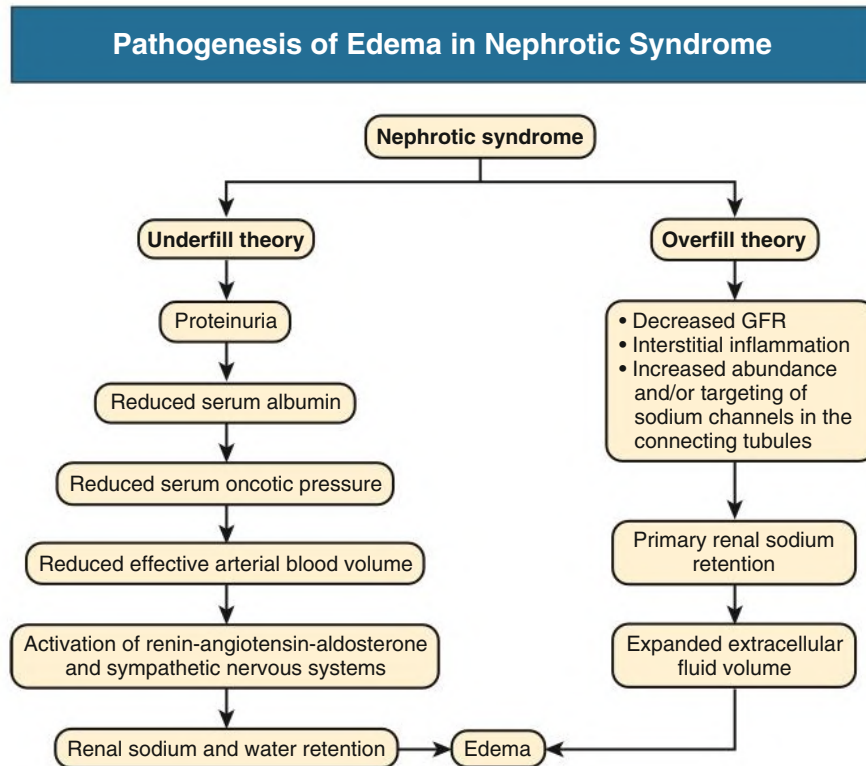
Many pathogenetic aspects of sodium and water retention are similar in cirrhosis and HF (Fig. 8.4). However, arterial underfilling in cirrhosis occurs secondary to splanchnic arterial vasodilation, with resultant water and sodium retention. Ascites formation in cirrhosis is likely initiated by sinusoidal and portal hypertension<sup>39</sup> from distortion of hepatic architecture, increased hepatic vascular tone, or increased splenohepatic flow. Decreased intrahepatic bioavailability of NO and increased production of vasoconstrictors (Ang II, endothelin) also are responsible for increased resistance in the hepatic vasculature.<sup>40</sup> Portal hypertension caused by increased sinusoidal pressure activates vasodilatory mechanisms in the splanchnic circulation.<sup>41</sup> These mechanisms, mediated at least partly by NO and carbon monoxide (CO) overproduction, lead to splanchnic and peripheral arteriolar vasodilation. In advanced stages of cirrhosis, arteriolar vasodilation causes underfilling of the systemic arterial vascular space. This event, through a decrease in EABV, leads to a fall in arterial pressure. Consequently, baroreceptor-mediated activation of the RAAS, SNS stimulation, and nonosmotic release of ADH occur to restore the blood volume homeostasis.<sup>42</sup> This involves compensatory vasoconstriction as well as kidney sodium and water retention. However, splanchnic vasodilation also increases splanchnic lymph production,

which exceeds the lymph-transporting capacity, and thus lymph leakage into the peritoneal cavity occurs with ascites development.<sup>43</sup> Persistent kidney sodium and water retention (along with lymph leakage into the peritoneal cavity from increased splanchnic vascular permeability) play the major role in sustained ascites formation.

### Sodium and Water Retention in Nephrotic Syndrome

Unlike HF and liver cirrhosis, in which the kidneys are structurally normal, the nephrotic syndrome is characterized by diseased kidneys. Many nephrotic patients have a higher arterial blood pressure, higher GFR, and less impairment of sodium and water excretion than patients with HF and cirrhosis. Two mechanisms, underfill and overfill, may account for nephrotic edema; components of each mechanism may exist in different edematous conditions (Fig. 8.5). The underfill theory suggests that reduced plasma oncotic pressure from proteinuria increases fluid movement from the vascular to interstitial compartment. The resultant arterial underfilling culminates in activation of homeostatic mechanisms involving the SNS and the RAAS. The overfill theory implicates primary kidney sodium and water retention that translates into elevated total plasma volume, hypertension, and suppressed RAAS. Distinguishing between the two mechanisms is important because it influences the approach to diuretic use in nephrotic patients.

The following observations support the underfill theory for edema formation. Plasma volume, systemic arterial blood pressure, and cardiac output are diminished in some nephrotic patients, especially in



**Fig. 8.5** Underfill and Overfill Mechanisms in Pathogenesis of Edema in Nephrotic Syndrome. The ratio of underfill and overflow likely varies depending on the cause of the nephrotic syndrome. *GFR*, Glomerular filtration rate.

children with minimal change disease (MCD; see Chapter 18). The Starling forces governing the fluid movement across the capillary wall equal the difference of the hydrostatic pressure and the oncotic pressure gradients. The gradual decrease in the plasma albumin concentration and the plasma oncotic pressure is mitigated by the reduced entry of albumin into the interstitial space and a concurrent decline in interstitial oncotic pressure. Consequently, less ECF volume expansion and edema formation is noted unless hypoalbuminemia is severe.<sup>44</sup> Thus, nephrotic patients who are underfilled and are predisposed to AKI despite generalized edema generally have serum albumin concentration of less than 2 g/dL (20 g/L). However, serum albumin concentration may not be the primary driver of ECF volume expansion because either spontaneous or steroid-induced remission of minimal change leads to natriuresis before albumin rises<sup>45</sup>; this suggests that other mechanisms, in addition to low oncotic pressure, must be involved.

Observations supporting the overfill theory include studies of adults with MCD who have increased blood volume and blood pressure. After remission induced by corticosteroids, plasma volume and blood pressure decline, with an increase in plasma renin activity. However, evaluation of intravascular volume is somewhat unreliable because the afferent stimulus for edema formation appears to be a dynamic process, with different results at different phases of edema formation.<sup>44</sup> Also supporting primary kidney sodium retention, experimental studies in animals with unilateral nephrotic syndrome demonstrate that sodium retention occurs secondary to increased reabsorption in the collecting tubules.<sup>46</sup> Increased abundance and apical targeting of ENaC subunits may be mediated, in part, by proteolytic cleavage of the channel by filtered proteases.<sup>47–49</sup>

In summary, nephrotic patients with arterial underfilling are more likely to have MCD with severe hypoalbuminemia, preserved GFR, and low blood pressure or postural hypotension. Other glomerular diseases, especially those involving kidney inflammation, are more often associated with an overfill picture with volume expansion, raised blood

pressure, and a decline in GFR. It has been postulated that interstitial inflammatory cells, a feature of some glomerular diseases other than MCD, may facilitate an increase in sodium retention and hypertension by releasing mediators that cause vasoconstriction.<sup>50</sup>

### Drug-Induced Edema

Although they are no longer commonly used, systemic vasodilators such as minoxidil and diazoxide induce arterial underfilling and subsequent retention of sodium with water (causing edema) through mechanisms similar to those in HF or cirrhosis. Dihydropyridine calcium channel blockers may cause peripheral edema, which is related to redistribution of fluid from the vascular space into the interstitium, possibly induced by preferential dilation of resistance arterioles in the absence of an appropriate microcirculatory myogenic reflex. This facilitates transmission of the systemic pressure to the capillary circulation.<sup>51</sup> Fluid retention and HF exacerbation may be seen with thiazolidinedione therapy in patients with type 2 diabetes mellitus, involving activation of peroxisome proliferator-activated receptor  $\gamma$  (PPAR $\gamma$ ). Although this can simulate sodium reabsorption by the sodium channels in collecting tubule cells,<sup>52</sup> additional effects are likely involved, including effects on vascular smooth muscle to reduce arterial pressure.<sup>53</sup> NSAIDs can exacerbate volume expansion in HF and cirrhotic patients by decreasing vasodilatory prostaglandins in the afferent arteriole of the glomerulus.<sup>54</sup>

### Idiopathic Edema

This poorly defined syndrome occurs most often in premenopausal women and is characterized by intermittent edema secondary to sodium and water retention. Patients often complain of face and hand edema, leg swelling, and variable weight gain<sup>55</sup> and often misuse diuretics or laxatives, which may chronically stimulate the RAAS. The diagnosis of idiopathic edema is usually made by exclusion of other causes after history, physical examination, and sometimes diuretic screening.

## Sodium and Water Retention in Pregnancy

In the first trimester of normal pregnancy, systemic arterial vasodilation and a decrease in blood pressure occur in association with a compensatory increase in cardiac output.<sup>56</sup> The lower arterial pressure activates the RAAS, contributing to the kidney sodium and water retention and expanding the plasma volume (10%–30%).<sup>57</sup> Decreased plasma osmolality, stimulated thirst, and persistent nonosmotic vasopressin release are other features of normal pregnancy. In contrast to diseases such as HF and cirrhosis, pregnancy is associated with an increase in GFR and kidney blood flow. The increased GFR leads to higher filtered load and increased distal sodium delivery in pregnancy, which contributes to the better escape from the sodium-retaining effect of aldosterone and attenuates edema formation. See [Chapter 44](#) for more details.

## Clinical Manifestations of Extracellular Fluid Volume Expansion

A history of an underlying disease, such as coronary artery disease, hypertension, or liver cirrhosis, can pinpoint the underlying mechanism of edema formation. Patients with left HF may present with exertional dyspnea, orthopnea, and paroxysmal nocturnal dyspnea. Patients with right-sided HF or biventricular failure may exhibit weight gain and lower limb swelling. Physical examination reveals increased JVP, pulmonary crackles, a third heart sound, or dependent peripheral edema that may be elicited in the ankles or sacrum.

Nephrotic patients may present with periorbital edema, especially when caused by minimal change disease. However, those with severe disease may exhibit marked generalized edema with anasarca. Patients with cirrhosis present with ascites and lower limb edema caused by portal hypertension and hypoalbuminemia. Physical examination may reveal stigmata of chronic liver disease and splenomegaly.

## Diagnostic and Therapeutic Approach to Extracellular Volume Expansion

Management of ECF volume expansion consists of recognizing and treating the underlying cause and attempting to achieve negative sodium balance by dietary sodium restriction and administration of diuretics. Before embarking on diuretic therapy, it is imperative to appreciate that ECF volume expansion may have occurred to compensate for arterial underfilling, as in HF and cirrhosis. A judicious approach is therefore necessary to avoid a precipitous fall in cardiac output and tissue perfusion. Nevertheless, it has become clear that diuresis of 3 to 5 L/day is typically tolerated safely in patients with acute decompensated HF, and that a modest rise in serum creatinine concentration should be accepted, as long as decongestion is taking place.<sup>58</sup>

Moderate dietary sodium restriction (2–3 g Na<sup>+</sup>/day; 86–130 mmol/day) should be encouraged. Salt substitutes contain potassium chloride and should not be used in patients with advanced kidney impairment or those taking potassium-sparing diuretics. Restriction of total fluid intake is usually necessary only for patients with hyponatremia. Medications that promote sodium retention (e.g., NSAIDs) should be discontinued. Diuretics are the cornerstone of therapy to remove excess volume. Other measures can be used in patients with inadequate response or lack of response to diuretics. In those with liver cirrhosis, large-volume paracentesis with albumin infusion provides symptomatic relief and is commonly included as part of the treatment regimen. Although extracorporeal fluid removal by ultrafiltration can be used in patients with acute decompensated HF accompanied by kidney impairment or diuretic resistance, a randomized controlled trial did not indicate benefit.<sup>59</sup>

Although salt and fluid removal is a central feature of the treatment of HF, there are a number of other approaches that have been shown

to ameliorate symptoms and prolong life. ACE inhibitors, angiotensin receptor blockers (ARBs),  $\beta$ -blockers, and mineralocorticoid receptor antagonists have been shown to be useful in those with HF<sub>REF</sub>. Recently, SGLT2 inhibitors, which have a modest diuretic effect, have also been shown to improve mortality in patients with HF.

## Diuretics

### Principles of Action

Diuretics are the mainstays of therapy for edematous states, with five classes based on the predominant sites of diuretic action along the nephron ([Fig. 8.6](#)). Most diuretics work from the tubule lumen and reach this site by secretion across the proximal tubule epithelium. All diuretics except osmotic agents have a high degree of protein binding, which limits glomerular filtration and traps them in vascular spaces, allowing them to be delivered to the proximal convoluted tubule for secretion.<sup>60</sup> Most diuretics either act by inhibiting sodium reabsorption with an accompanying anion, usually chloride (loop diuretics or thiazides), or inhibit sodium reabsorption through channels in the luminal membrane (amiloride directly or aldosterone antagonists, indirectly). The resultant natriuresis decreases the ECF volume.

Loop diuretics, which are used commonly to treat edematous conditions, have relatively short half-lives, and each dose is followed by a phase of natriuresis and a phase of antinatriuresis (often referred to as postdiuretic Na<sup>+</sup> retention). A low-salt diet can ensure that postdiuretic Na<sup>+</sup> retention does not overcome effective diuresis. A second type of adaptation occurs during chronic treatment. During continued administration of a diuretic, the magnitude of each natriuretic response declines, so daily salt excretion once again equals daily salt intake. This is known as the *braking phenomenon*; mechanisms include activation of the SNS and RAAS, decreased systemic and kidney arterial blood pressure, hypertrophy of the distal nephron cells with increased expression of epithelial transporters, and perhaps alterations in natriuretic hormones (e.g., ANP).<sup>60</sup>

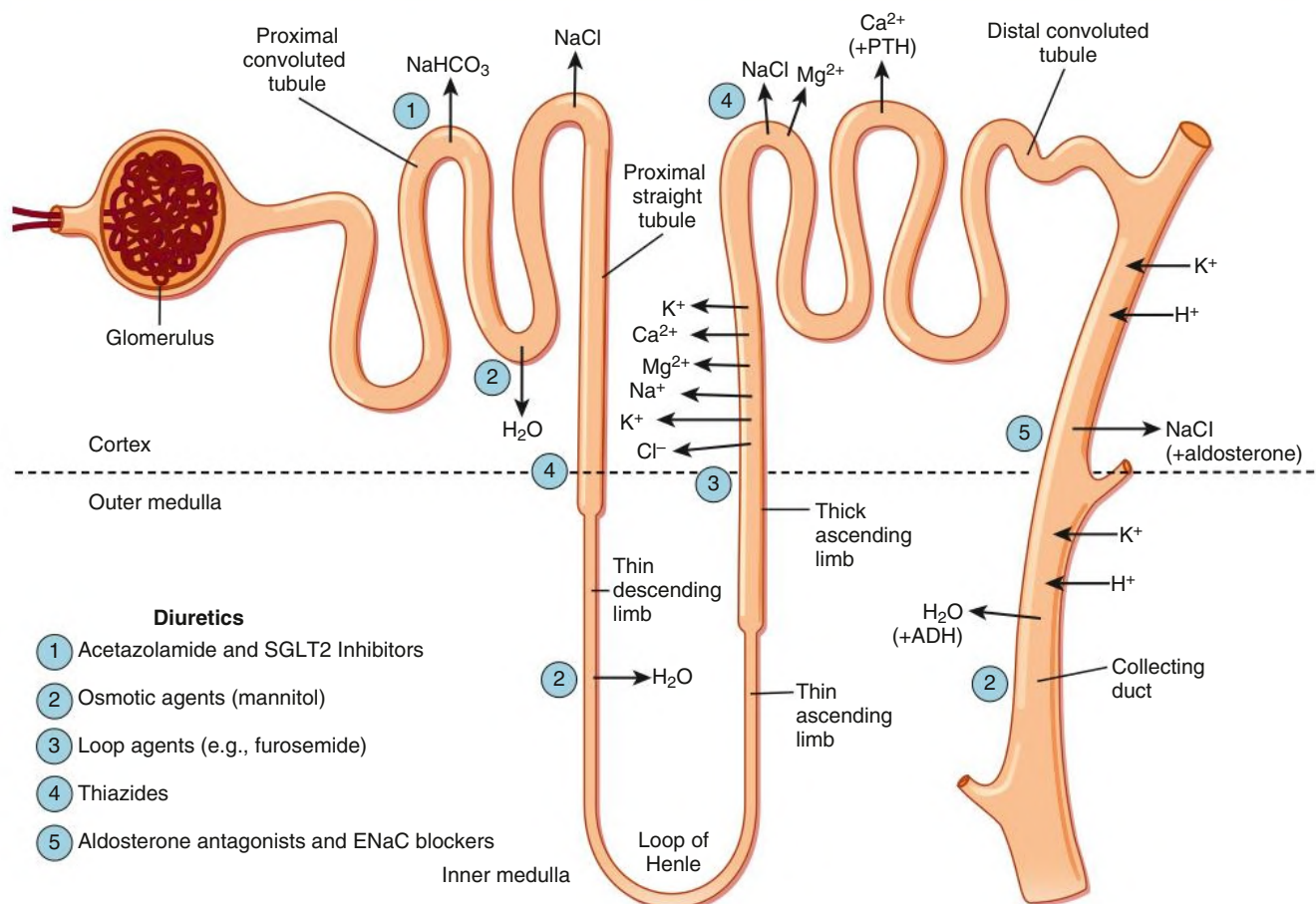
### Classes of Diuretics

**Loop diuretics.** Loop diuretics such as furosemide, bumetanide, and torsemide act by blocking the Na<sup>+</sup>-K<sup>+</sup>-Cl<sup>-</sup> cotransporters at the luminal surface of thick ascending limb (TAL) cells, thereby diminishing net reabsorption.<sup>61</sup> These agents are the most potent of all diuretics, inhibiting the reabsorption of 25% of filtered sodium, which normally occurs along the TAL. Moreover, the nephron segments past the TAL typically do not possess the capacity to reabsorb completely the volume of fluid exiting the TAL. Thus, a substantial percentage of the solute rejected from the TAL is excreted.

The oral bioavailability of furosemide averages 50%, but varies between 10% and 100%; the oral bioavailabilities of bumetanide and torsemide are higher (~80%). Loop diuretics have short elimination half-lives, so the dosing interval needs to be short to maintain adequate levels in the lumen. Excessive prolongation of dosing interval provides more time for postdiuretic sodium retention to overcome natriuresis. This is the reason that loop diuretics are typically given twice daily. When escalating diuretics in congested hospitalized patients, several algorithms recommend repeating furosemide doses as frequently as every 6 hours, given the short intravenous (IV) half-life of furosemide and bumetanide.<sup>62</sup> For oral use, torsemide has a longer half-life than bumetanide or furosemide, permitting it to be used once daily in some cases.

The intrinsic potency of a diuretic is defined by its dose-response curve, which is generally sigmoidal. The steep dose-response is the reason that loop diuretics are often referred to as *threshold drugs*. This is exemplified by furosemide, which can initiate diuresis in a person with normal kidney function with an IV dose of 10 mg, and a maximal effect

## Sites of Salt and Water Transport and Actions of Diuretics



**Fig. 8.6** Sites of Salt and Water Transport and Actions of Diuretics. Specific sites are indicated by numbers. (Modified from Ives HE. Diuretic agents. In: Katzung BG, ed. *Basic and Clinical Pharmacology*. McGraw-Hill Medical; 2006:237.)

is seen with 40 mg. A larger dose provides minimal or no extra increase in natriuresis, and side effects may increase. In contrast, the effective diuretic dose is often substantially higher in patients with HF, advanced cirrhosis, and kidney failure; guidelines are available for diuretic dosing in these situations.<sup>62</sup> Patients who respond poorly to intermittent doses of a loop diuretic may receive continuous IV infusion, which might enhance the response by virtue of maintaining an effective amount of drug at the site of action.<sup>63</sup> A Cochrane review suggested that continuous infusions led to greater diuresis and a better safety profile, but the quality of the evidence is poor.<sup>64</sup> A randomized trial in acute decompensated HF found no difference in the primary end points—the global assessment of symptoms over the course of 72 hours and the change in serum creatinine from baseline to 72 hours—for continuous versus bolus infusion of furosemide. A higher dose of diuretic was more effective without clinically important negative effects on kidney function.<sup>65</sup> Despite this negative result, this study did not include diuretic-resistant patients or those with significant kidney dysfunction, and so continuous infusion is still employed in some cases.

Although pharmacologically similar to other loop diuretics, ethacrynic acid has greater ototoxic potential and therefore is reserved for patients allergic to other agents.

**Distal convoluted tubule diuretics.** Thiazide and thiazide-like diuretics (chlorothiazide, hydrochlorothiazide, chlorthalidone,

metolazone, and indapamide) inhibit NaCl absorption in the distal convoluted tubule, where up to 5% of filtered sodium and chloride is reabsorbed, and are therefore less potent than loop diuretics. Thiazides have a relatively longer half-life and can be administered once or twice daily. Metolazone has pharmacologic characteristics similar to classical thiazide diuretics and is more often used in conjunction with other classes of diuretics.

Thiazides are used commonly to treat hypertension (see [Chapter 37](#)). For edema, they are occasionally used alone in patients with mild HF, but more often they are used in combination to synergize the effect of loop diuretics by blocking multiple nephron segment sites. Because thiazide diuretics must reach the lumen to be effective, higher doses are required in patients with impaired kidney function. Thiazides as sole agents (possibly excluding the thiazide-like diuretics metolazone and indapamide) lose some effectiveness at GFR less than 30 mL/min but may still be useful in some patients.<sup>66</sup> Regardless, thiazide diuretics still enhance the diuretic effect of loop diuretics when coadministered with loop diuretics to overcome diuretic resistance. If used, such combination therapy should be closely monitored because of a risk for hypokalemia and excessive ECF depletion.

**Collecting duct diuretics.** Amiloride, triamterene, and the mineralocorticoid receptor antagonists spironolactone and eplerenone act on the collecting duct. Amiloride and triamterene act primarily in



the connecting tubule and cortical collecting duct (the aldosterone-sensitive distal nephron) by interfering with sodium reabsorption through the apical sodium channel ENaC; this inhibits potassium secretion by dissipating the electronegative gradient (normally created by sodium reabsorption) that favors potassium secretion. Spironolactone and eplerenone are competitive antagonists of aldosterone and cause natriuresis and potassium retention. Finerenone is a new nonsteroidal mineralocorticoid receptor antagonist, which has properties that appear to differ from those of spironolactone and eplerenone; it appears to block the mineralocorticoid receptor as a bulky, passive agonist. It was recently approved by the US Food and Drug Administration (FDA) to slow progression of CKD in the setting of diabetes mellitus.<sup>67</sup> Potassium-sparing diuretics are considered weak diuretics because they block only about 3% of the filtered sodium load reaching their site of action and thus are most often used with other diuretics to augment diuresis or preserve potassium. If combination therapy is used, careful monitoring is essential to prevent dangerous hyperkalemia. Vulnerable patients include those with underlying kidney dysfunction, those with HF, patients with diabetes, and those concurrently taking ACE inhibitors, ARBs, NSAIDs, or  $\beta$ -blockers.

Collecting duct diuretics are considered first-line agents in certain conditions. For example, spironolactone is used in patients with liver cirrhosis with ascites and amiloride in the treatment of Liddle syndrome, a rare autosomal dominant condition characterized by a primary increase in ENaC function (see [Chapter 47](#)).

**Proximal tubule diuretics.** A substantial component of sodium reabsorption along the proximal tubule is mediated by a luminal  $\text{Na}^+\text{-H}^+$  exchanger. The activity of this transporter depends on both the generation of protons inside the cell and their elimination in the luminal brush border. Both actions are catalyzed by carbonic anhydrase. Acetazolamide inhibits carbonic anhydrase, thereby reducing proton secretion in the proximal tubule lumen, thus increasing sodium bicarbonate excretion. Acetazolamide is a weak diuretic because distal segments reabsorb much of the inhibited  $\text{Na}^+$  and because proximal inhibition activates the TGF to reduce  $\text{Na}^+$  filtration.<sup>8</sup> Acetazolamide generates hyperchloremic metabolic acidosis, particularly with prolonged use. It also may cause hypokalemia because of increased distal sodium delivery; it may cause hypophosphatemia, but the mechanism of this is not well understood. Rarely used as a single agent, acetazolamide is most frequently used with other diuretics, in treatment of metabolic alkalosis accompanied by edematous states, and in chronic obstructive pulmonary disease. It is also used in acute and chronic high-altitude sickness because of its ability to stimulate ventilation as a compensatory response to the metabolic acidosis, although unpleasant paresthesias may occur.

**Osmotic diuretics.** Osmotic diuretics such as mannitol are inert substances that are freely filtered at the glomerulus but are poorly reabsorbed. Mannitol is used intravenously and produces diuresis by increasing the osmotic pressure within the lumen of the proximal tubule and loop of Henle. This causes enhanced water diuresis and to a lesser extent sodium and potassium excretion. Mannitol is not used for edematous states but rather to treat cerebral edema induced by trauma or neoplasms and to reduce intraocular pressure. Mannitol is also used in the treatment of dialysis disequilibrium syndrome, preventing the brisk decline in serum osmolality that would otherwise occur with dialysis. This helps prevent a shift of fluid into the brain that may occur with rapid rate of solute removal by dialysis, which is thought to be responsible for the symptoms.

**Aquaretics.** The “vaptans” are drugs that block either  $V_2$  receptors alone or  $V_1$  and  $V_2$  receptors together and increase electrolyte-free water excretion. They are used in electrolyte disorders, such as the syndrome of inappropriate antidiuretic hormone secretion, but have

also been approved to slow CKD progression in autosomal dominant polycystic kidney disease. Because they have little effect on urinary sodium excretion, they are not typically effective in addressing edema.

**SGLT2 inhibitors.** SGLT2 inhibitors, initially introduced as drugs to control hyperglycemia, have pleiotropic effects that improve kidney and cardiovascular outcomes, even in the absence of diabetes mellitus. Although not marketed as diuretics, these agents cause a mild diuresis, both by inhibiting sodium glucose cotransport along the proximal tubule and by inhibiting the  $\text{Na}^+\text{-H}^+$  exchanger. Sodium glucose cotransport and sodium proton antiport appear to be coupled functionally along the proximal tubule. These drugs induce a mild but sustained reduction in ECF volume that may contribute to their beneficial effects.<sup>68</sup>

### Adverse Effects

Many common diuretics are derived from sulfanilamide and may therefore induce allergy in susceptible patients, manifested as hypersensitivity reactions, usually as a rash or, rarely, acute interstitial nephritis. The most serious and common adverse effects of diuretics, however, are electrolyte disturbances.

Loop diuretics impair tubular reabsorption by inhibiting the  $\text{Na/K/2Cl}$  cotransporter along the luminal membrane of thick ascending limb cell. This transporter lies in parallel with a potassium channel that permits potassium to diffuse back into the lumen. Coupled with basolateral chloride exit, this drives an electrical potential difference oriented with the lumen positive with respect to blood. This drives magnesium and calcium reabsorption. Loop diuretics eliminate this potential difference and can, therefore, increase calcium and magnesium excretion. Thiazide diuretics exert the same effect on magnesium, although the mechanism is different. In a recent study of community-dwelling subjects, the use of a thiazide was associated with prevalent hypomagnesemia, whereas loop diuretic use was not.<sup>69</sup> Unlike loop diuretics, thiazide diuretics decrease urinary calcium losses and are therefore preferred in the treatment of hypercalciuric states and in patients with osteoporosis. Thiazide diuretics interfere with urine-diluting mechanisms by blocking sodium reabsorption at the distal convoluted tubule, an effect that may contribute to hyponatremia.<sup>70</sup> Acutely, loop and thiazide diuretics increase the excretion of uric acid, whereas chronic administration results in reduced uric acid excretion. The chronic effect is likely caused by enhanced absorption in the proximal convoluted tubule secondary to volume depletion. Other adverse effects with large doses may include ototoxicity with loop diuretics, particularly with aminoglycoside coadministration, and gynecomastia, which may develop with spironolactone.

### Approach to Diuretic Treatment of Extracellular Fluid Volume Expansion

Although the mechanisms underlying expansion of the ECF volume are similar in the different clinical syndromes, some evidence-based differences in diuretic approach are recommended. For HF, either  $\text{HF}_{\text{REF}}$  or  $\text{HF}_{\text{pEF}}$ , initial treatment involves loop diuretics. A starting dose of 20 to 40 mg furosemide, or an equivalent, is typical, and may be effective, but it may be necessary to increase, or double, subsequent doses if the initial dose does not elicit a diuresis. Such a diuresis should occur promptly after drug administration and be notable to the patient. This approach is based on the threshold nature of the loop diuretic dose response curve. It is often necessary to use loop diuretics twice daily, but it is always important to ensure that each dose is above the threshold. The pharmacokinetic profile of torsemide is more favorable than that of furosemide; some clinicians prefer this drug, but solid evidence to support this choice is lacking currently.

For cirrhotic ascites, patients are typically started on a combination of spironolactone and furosemide, at a fixed ratio (100 mg spironolactone to 40 mg furosemide).<sup>71</sup> Single-agent spironolactone can be used but causes more potassium imbalance. Doses can be titrated retaining the same ratio of spironolactone and furosemide.

For nephrotic syndrome and CKD, loop diuretics are indicated, but higher doses than used for HF are typically required because both CKD and nephrotic syndrome are diuretic-resistant conditions.

### Diuretic Resistance

Diuretic *resistance* typically refers to edema that has become refractory to maximal doses of loop diuretics. Apparent resistance may be the result of drug interactions. NSAIDs block prostaglandin-mediated increases in kidney blood flow and natriuresis and can

precipitate HF exacerbations.<sup>54</sup> Arterial underfilling in cirrhosis and HF increases proximal tubular sodium reabsorption, which reduces delivery of sodium to the distal nephron segment sites of diuretic action. Yet in HF, resistance most commonly results from activation of distal nephron transport.<sup>72</sup> This problem can be addressed by combining loop and thiazide diuretics because the latter block the distal nephron sites responsible. To prevent loop diuretic-induced renin-angiotensin-aldosterone effects, many patients are treated with aldosterone antagonists, especially because these agents are essential parts of the treatment regimen for HFrEF. If other approaches fail, a thiazide diuretic can be added. In nephrotic syndrome, the filtered protein in tubular fluid may contain serine proteases, which cleave and activate ENaC, thereby promoting excess sodium reabsorption.<sup>49</sup> ENaC can be inhibited with amiloride.

## SELF-ASSESSMENT QUESTIONS

1. A 58-year-old woman with a history of alcohol dependence and known chronic liver disease presents to the emergency department with increasing breathlessness and distended abdomen. Vital signs reveal temperature of 36.9°C (98.5°F), respiratory rate of 17 breaths/min, oxygen saturation of 93% on room air, pulse rate of 76 beats/min, and blood pressure of 93/58 mm Hg without significant orthostatic change. Physical examination reveals scleral icterus, and JVP is flat. Cardiac auscultation is normal; breath sounds are decreased at the bases. Her abdomen is distended with a fluid wave and a positive shifting dullness. The patient has bilateral pitting edema to the mid-calves and intact peripheral pulses. Which of the following is the primary mechanism responsible for *maintaining* volume excess in this patient?
  - A. Reduction in plasma oncotic pressure resulting from synthetic hypoalbuminemia causing an increase in fluid movement from the vascular to the interstitial compartment.
  - B. Underfilling of the arterial circulation resulting from splanchnic venous dilation causing neurally and hormonally mediated kidney sodium retention.
  - C. Primary kidney sodium and water retention that results in elevated total plasma volume.
  - D. Decrease in plasma osmolality with stimulation of thirst and persistent release of antidiuretic hormone.
2. Laboratory investigations of the patient described in question 1 reveal serum sodium of 128 mmol/L, potassium of 3.7 mmol/L, chloride of 101 mmol/L, bicarbonate of 23 mg/dL, BUN of 56 mg/dL, and creatinine of 1.0 mg/dL. Albumin is 3.0 g/dL. Reasonable initial approaches to remove excess volume in this patient include all of the following *except*:
  - A. Prescribe a diet restricted to 2.3 g (100 mmol) of sodium daily to mitigate kidney retention of sodium and prevent diuretic resistance.
  - B. Perform a large-volume paracentesis with albumin infusion to rapidly decompress the abdomen.
  - C. Administer spironolactone up to 400 mg/day, with a loop diuretic to increase sodium excretion.
  - D. Administer intravenous mannitol to increase osmotic pressure within the tubule and thus sodium and water reabsorption.
3. Over the next 3 days, the patient previously described becomes febrile, with a temperature of 37.8°C (100°F), and confused. Her pulse rate is 120 beats/min with blood pressure of 88/52 mm Hg. The JVP is now near the angle of the jaw when the patient is sitting at 45 degrees. Hemoglobin is 7.6 g/dL, total white blood cell (WBC) count 15,000 (85% polymorphonuclear leukocytes), and platelet count  $81,000 \times 10^9$  cells/L. Analysis of the ascitic fluid reveals 100 RBCs/ $\mu$ L and 450 WBCs/ $\mu$ L, but no organisms on Gram stain. BUN is 48 mg/dL, serum creatinine is 1.7 mg/dL, and urine sodium is 65 mEq/L, with a fractional sodium excretion of 1.4%. Appropriate immediate measures for the patient at this time would include which of the following?
  - A. Administer 50 g of albumin, with the goal of improving her hemodynamic profile.
  - B. Administer fluid resuscitation with 6% hydroxyethyl starch to expand extracellular fluid compartment rapidly and restore blood pressure.
  - C. Treat for presumptive bacterial peritonitis.
  - D. Insert jugular venous dialysis access for urgent initiation of dialysis.

## REFERENCES

1. Titze J. A different view on sodium balance. *Curr Opin Nephrol Hypertens*. 2015;24(1):14–20.
2. Titze J. Water-free sodium accumulation. *Semin Dial*. 2009;22(3):253–255.
3. Jantsch J, Schatz V, Friedrich D, et al. Cutaneous Na<sup>+</sup> storage strengthens the antimicrobial barrier function of the skin and boosts macrophage-driven host defense. *Cell Metab*. 2015;21(3):493–501.
4. Rakova N, Juttner K, Dahlmann A, et al. Long-term space flight simulation reveals infradian rhythmicity in human Na(+) balance. *Cell Metab*. 2013;17(1):125–131.
5. Maroudas AI. Balance between swelling pressure and collagen tension in normal and degenerate cartilage. *Nature*. 1976;260(5554):808–809.
6. Guyton AC, Coleman TG, Granger HJ. Circulation: overall regulation. *Annu Rev Physiol*. 1972;34:13–46.
7. Miller WL, Mullan BP. Understanding the heterogeneity in volume overload and fluid distribution in decompensated heart failure is key to optimal volume management: role for blood volume quantitation. *JACC Heart Fail*. 2014;2(3):298–305.
8. Palmer LG, Schnermann J. Integrated control of Na transport along the nephron. *Clin J Am Soc Nephrol*. 2015;10(4):676–687.
9. Salah HM, Al'Aref SJ, Khan MS, et al. Effect of sodium-glucose cotransporter 2 inhibitors on cardiovascular and kidney outcomes-Systematic review and meta-analysis of randomized placebo-controlled trials. *Am Heart J*. 2021;232:10–22.
10. Anders HJ, Davis JM, Thurau K. Nephron protection in diabetic kidney disease. *N Engl J Med*. 2016;375:2096–2098.
11. Redfield MM, Edwards BS, Heublein DM, Burnett Jr JC. Restoration of renal response to atrial natriuretic factor in experimental low-output heart failure. *Am J Physiol*. 1989;257:R917–R923.
12. Hall JE, Granger JP, Smith Jr MJ, Premen AJ. Role of renal hemodynamics and arterial pressure in aldosterone “escape”. *Hypertension*. 1984;6(2 Pt 2):1183–1192.
13. Hall JE, Granger JP, Hester RL, et al. Mechanisms of escape from sodium retention during angiotensin II hypertension. *Am J Physiol*. 1984;246(5 Pt 2):F627–F634.
14. Anders HJ, Davis JM, Thurau K. Nephron protection in diabetic kidney disease. *N Engl J Med*. 2016;375(21):2096–2098.
15. DiBona GF. Physiology in perspective: the Wisdom of the Body. Neural control of the kidney. *Am J Physiol Regul Integr Comp Physiol*. 2005;289(3):R633–R641.
16. Akabane S, Matsushima Y, Matsuo H, et al. Effects of brain natriuretic peptide on renin secretion in normal and hypertonic saline-infused kidney. *Eur J Pharmacol*. 1991;198(2–3):143–148.
17. Schrier RW, Berl T, Anderson RJ. Osmotic and nonosmotic control of vasopressin release. *Am J Physiol*. 1979;236(4):F321–F332.
18. Goldsmith SR. Vasopressin as a vasopressor. *Am J Med*. 1987;82:1213–1219.
19. Schafer JA, Hawk CT. Regulation of Na<sup>+</sup> channels in the cortical collecting duct by AVP and mineralocorticoids. *Kidney Int*. 1992;41:255–268.
20. Konishi KI, Mizuochi T, Yanagi T, et al. Clinical features, molecular genetics, and long-term outcome in congenital chloride diarrhea: a nationwide study in Japan. *J Pediatr*. 2019;214:151–157.e6.
21. Perazella MA, Coca SG. Traditional urinary biomarkers in the assessment of hospital-acquired AKI. *Clin J Am Soc Nephrol*. 2012;7:167–174.
22. Finfer S, Bellomo R, Boyce N, et al. A comparison of albumin and saline for fluid resuscitation in the intensive care unit. *N Engl J Med*. 2004;350(22):2247–2256.
23. Perner A, Haase N, Guttormsen AB, et al. Hydroxyethyl starch 130/0.42 versus Ringer’s acetate in severe sepsis. *N Engl J Med*. 2012;367(2):124–134.
24. Reinhart K, Perner A, Sprung CL, et al. Consensus statement of the ESICM task force on colloid volume therapy in critically ill patients. *Intensive Care Med*. 2012;38(3):368–383.
25. Young P, Bailey M, Beasley R, et al. Effect of a buffered crystalloid solution vs saline on acute kidney injury among patients in the intensive care unit: the SPLIT randomized clinical trial. *JAMA*. 2015;314(16):1701–1710.
26. Semler MW, Self WH, Wanderer JP, et al. Balanced crystalloids versus saline in critically ill adults. *N Engl J Med*. 2018;378:829–839.
27. Hinrichs GR, Jensen BL, Svenningsen P. Mechanisms of sodium retention in nephrotic syndrome. *Curr Opin Nephrol Hypertens*. 2020;29:207–212.
28. Wang XY, Masilamani S, Nielsen J, et al. The renal thiazide-sensitive Na-Cl cotransporter as mediator of the aldosterone-escape phenomenon. *J Clin Invest*. 2001;108(2):215–222.
29. Yokota N, Bruneau BG, Kuroski de Bold ML, de Bold AJ. Atrial natriuretic factor significantly contributes to the mineralocorticoid escape phenomenon. Evidence for a guanylate cyclase-mediated pathway. *J Clin Invest*. 1994;94(5):1938–1946.
30. Schrier RW. Body fluid volume regulation in health and disease: a unifying hypothesis. *Ann Intern Med*. 1990;113:155–159.
31. Buglioni A, Burnett Jr JC. Pathophysiology and the cardiorenal connection in heart failure. Circulating hormones: biomarkers or mediators. *Clin Chim Acta*. 2015;443:3–8.
32. Sparks MA, Stegbauer J, Chen D, et al. Vascular type 1A angiotensin II receptors control BP by regulating renal blood flow and urinary sodium excretion. *J Am Soc Nephrol*. 2015;26(12):2953–2962.
33. Khraibi AA, Granger JP, Haas JA, et al. Intrarenal pressures during direct inhibition of sodium transport. *Am J Physiol Regul Integr Comp Physiol*. 1992;263:R1182–R1186.
34. Hensen J, Abraham WT, Durr JA, Schrier RW. Aldosterone in congestive heart failure: analysis of determinants and role in sodium retention. *Am J Nephrol*. 1991;11(6):441–446.
35. Schrier RW, Cadnapaphornchai MA, Ohara M. Water retention and aquaporins in heart failure, liver disease and pregnancy. *J R Soc Med*. 2001;94(6):265–269.
36. Szatalowicz VL, Arnold PE, Chaimovitz C, et al. Radioimmunoassay of plasma arginine vasopressin in hyponatremic patients with congestive heart failure. *N Engl J Med*. 1981;305:263–266.
37. Birnbaumer M, Seibold A, Gilbert S, et al. Molecular cloning for the receptor for human antidiuretic hormone. *Nature*. 1992;357:333–335.
38. Lee WH, Packer M. Prognostic importance of serum sodium concentration and its modification by converting-enzyme inhibition in patients with severe chronic heart failure. *Circulation*. 1986;73(2):257–267.
39. Gines P, Schrier RW. Renal failure in cirrhosis. *N Engl J Med*. 2009;361(13):1279–1290.
40. Hernandez-Guerra M, Garcia-Pagan JC, Bosch J. Increased hepatic resistance: a new target in the pharmacologic therapy of portal hypertension. *J Clin Gastroenterol*. 2005;39(4 suppl 2):S131–S137.
41. Gines P, Cardenas A, Arroyo V, Rodes J. Management of cirrhosis and ascites. *N Engl J Med*. 2004;350(16):1646–1654.
42. Schrier RW, Arroyo V, Bernardi M, et al. Peripheral arterial vasodilation hypothesis: a proposal for the initiation of renal sodium and water retention in cirrhosis. *Hepatology*. 1988;8:1151–1157.
43. Arroyo V, Gines P. Mechanism of sodium retention and ascites formation in cirrhosis. *J Hepatol*. 1993;17(suppl 2):S24–S28.
44. Koomans HA, Geers AB, Dorhout Mees EJ, Kortlandt W. Lowered tissue-fluid oncotic pressure protects the blood volume in the nephrotic syndrome. *Nephron*. 1986;42(4):317–322.
45. Oliver WJ, Owings CL. Sodium excretion in the nephrotic syndrome: relation to serum albumin concentration, glomerular filtration rate, and aldosterone secretion rate. *Am J Dis Child*. 1967;113:352–362.
46. Ichikawa I, Rennke HG, Hoyer JR, et al. Role for intrarenal mechanisms in the impaired salt excretion of experimental nephrotic syndrome. *J Clin Invest*. 1983;71:91–103.
47. Andersen H, Friis UG, Hansen PB, et al. Diabetic nephropathy is associated with increased urine excretion of proteases plasmin, prostatic and urokinase and activation of amiloride-sensitive current in collecting duct cells. *Nephrol Dial Transplant*. 2015;30(5):781–789.
48. Svenningsen P, Bistrup C, Friis UG, et al. Plasmin in nephrotic urine activates the epithelial sodium channel. *J Am Soc Nephrol*. 2009;20(2):299–310.

49. Svenningsen P, Friis UG, Versland JB, et al. Mechanisms of renal NaCl retention in proteinuric disease. *Acta Physiol (Oxf)*. 2013;207(3):536–545.
50. Rodriguez-Iturbe B, Herrera-Acosta J, Johnson RJ. Interstitial inflammation, sodium retention, and the pathogenesis of nephrotic edema: a unifying hypothesis. *Kidney Int*. 2002;62(4):1379–1384.
51. Gustafsson D. Microvascular mechanisms involved in calcium antagonist edema formation. *J Cardiovasc Pharmacol*. 1987;10(suppl 1):S121–S131.
52. Guan Y, Hao C, Cha DR, et al. Thiazolidinediones expand body fluid volume through PPARgamma stimulation of ENaC-mediated renal salt absorption. *Nat Med*. 2005;11(8):861–866.
53. Kleinhenz JM, Murphy TC, Pokutta-Paskaleva AP, et al. Smooth muscle-targeted overexpression of peroxisome proliferator activated receptor-gamma disrupts vascular wall structure and function. *PLoS One*. 2015;10(10):e0139756.
54. Heerdink ER, Leufkens HG, Herrings RMC, et al. NSAIDs associated with increased risk of congestive heart failure in elderly patients taking diuretics. *Arch Intern Med*. 1998;158:1108–1112.
55. Streeten DH. Idiopathic edema. Pathogenesis, clinical features, and treatment. *Endocrinol Metab Clin North Am*. 1995;24(3):531–547.
56. Chapman AB, Abraham WT, Zamudio S, et al. Temporal relationships between hormonal and hemodynamic changes in early human pregnancy. *Kidney Int*. 1998;54(6):2056–2063.
57. West CA, Sasser JM, Baylis C. The enigma of continual plasma volume expansion in pregnancy: critical role of the renin-angiotensin-aldosterone system. *Am J Physiol Renal Physiol*. 2016;311(6):F1125–F1134.
58. Parikh CR, Coca SG. “Permissive AKI” with treatment of heart failure. *Kidney Int*. 2019;96:1066–1068.
59. Bart BA, Goldsmith SR, Lee KL, et al. Ultrafiltration in decompensated heart failure with cardiorenal syndrome. *N Engl J Med*. 2012;367(24):2296–2304.
60. Ellison DH, Hoorn EJ, Wilcox CS. Diuretics. In: Taal MW, Chertow GM, Marsden PA, et al., eds. *Brenner and Rector’s the Kidney*. 9th ed. Vol. 2. Philadelphia: Elsevier; 2012:1879–1916.
61. Schlatter E, Greger R, Weidtko C. Effect of ‘high ceiling’ diuretics on active salt transport in the cortical thick ascending limb of Henle’s loop of rabbit kidney. *Pflugers Arch*. 1983;396:210–217.
62. Felker GM, Ellison DH, Mullens W, Cox ZL, Testani JM. Diuretic therapy for patients with heart failure: JACC state-of-the-art review. *J Am Coll Cardiol*. 2020;75(10):1178–1195.
63. Rudy DW, Voelker JR, Greene PK, et al. Loop diuretics for chronic renal insufficiency: a continuous infusion is more efficacious than bolus therapy. *Ann Intern Med*. 1991;115:360–366.
64. Salvador DR, Rey NR, Ramos GC, Punzalan FE. Continuous infusion versus bolus injection of loop diuretics in congestive heart failure. *Cochrane Database Syst Rev*. 2005;3:CD003178.
65. Felker GM, Lee KL, Bull DA, et al. Diuretic strategies in patients with acute decompensated heart failure. *N Engl J Med*. 2011;364(9):797–805.
66. Sinha AD, Agarwal R. Thiazide diuretics in chronic kidney disease. *Curr Hypertens Rep*. 2015;17:13.
67. Bakris GL, Agarwal R, Anker SD, et al. Effect of finerenone on chronic kidney disease outcomes in type 2 diabetes. *N Engl J Med*. 2020;383:2219–2229.
68. Vallon V, Verma S. Effects of SGLT2 inhibitors on kidney and cardiovascular function. *Annu Rev Physiol*. 2021;83:503–528.
69. Liamis G, Rodenburg EM, Hofman A, et al. Electrolyte disorders in community subjects: prevalence and risk factors. *Am J Med*. 2013;126(3):256–263.
70. Leung AA, Wright A, Pazo V, et al. Risk of thiazide-induced hyponatremia in patients with hypertension. *Am J Med*. 2011;124(11):1064–1072.
71. Runyon BA, American Association for the Study of Liver Diseases, Practice Guidelines Committee. Management of adult patients with ascites due to cirrhosis: an update. *Hepatology*. 2009;49(6):2087–2107.
72. Ter Maaten JM, Rao VS, Hanberg JS, et al. Renal tubular resistance is the primary driver for loop diuretic resistance in acute heart failure. *Eur J Heart Fail*. 2017;19(8):1014–1022.
73. Ginés P, Cardenas A, Sola E, Schrier RW. Liver disease and the kidney. In: Coffman TM, Falk RJ, Molitoris BA, et al., eds. *Schrier’s Diseases of the Kidney*. Philadelphia: Wolters Kluwer; 2012:1965–1996.
74. Ives HE. Diuretic agents. In: Katzung BG, ed. *Basic and Clinical Pharmacology*. Columbus: McGraw-Hill Medical; 2006:237.